RESPIRATORY SYSTEM

NCLEX STUDY GUIDE

Assessment • Oxygenation • Ventilation • Emergencies

1. Assessment and Diagnostic Procedures

- Perform respiratory assessment in this order: inspection, palpation, percussion and auscultation.
- Sputum specimen: collect early morning after rinsing the mouth with water. Use a sterile container; deep expectoration or suction may be required. Three specimens are commonly collected when evaluating tuberculosis.

Procedure	Before	After / complication
Laryngoscopy or bronchoscopy	Consent, NPO, coagulation studies; remove dentures/eyeglasses; have suction and resuscitation equipment available.	Check gag reflex before oral intake. Watch for pneumothorax: unilateral absent breath sounds, dyspnea or crepitus.
Thoracentesis	Consent, chest imaging/coagulation studies; sit upright leaning over bedside table.	Monitor for pneumothorax, bleeding, embolism and re-expansion pulmonary edema.
Pulmonary function tests	Loose clothing; no heavy meal/smoking 4-6 hr; hold bronchodilators only as ordered.	Evaluates mechanics, volumes, gas exchange and acid-base/ABG patterns.

PRIORITY After an invasive chest procedure, sudden dyspnea, unilateral decreased breath sounds or subcutaneous crepitus requires immediate action.

2. Chest Injury and Mechanical Ventilation

Injury	Key findings / priorities
Rib fracture	Localized pain worse with inspiration, tenderness and shallow respirations. Fowler position, analgesia, pulmonary hygiene; monitor for pneumothorax.
Flail chest	At least 2 adjacent ribs fractured in multiple places; paradoxical chest motion, severe pain, hypoxemia and hypotension. Oxygen, ventilatory support and ICU-level monitoring.
Pneumothorax	Unilateral absent/decreased breath sounds, crepitus, chest pain, tachycardia and dyspnea. Tension pneumothorax may cause tracheal deviation away from affected side.
Open pneumothorax	Sucking chest wound. Apply a sterile occlusive dressing taped on three sides and prepare for chest tube.

Mode	Meaning
Control	Ventilator supplies breaths for a client unable to initiate respiratory effort.
Assist-control	Client may trigger a breath; ventilator delivers the preset breath. Hyperventilation/respiratory alkalosis can occur.
SIMV	Mandatory breaths synchronized with spontaneous breaths; often used during weaning.

Low-pressure alarm: leak/disconnection	High-pressure alarm: resistance/obstruction
Disconnected tubing, circuit/cuff leak, power or ventilator problem, reduced spontaneous breathing.	Secretions, bronchospasm/wheezing, coughing/gagging, biting tube, fighting ventilator, displaced/kinked tube.

VENTILATOR ALARM Assess the client first. If ventilation is inadequate, disconnect from the ventilator and manually ventilate with a bag-valve device while the cause is corrected.

3. Chest Tubes

Chamber	Expected / action
Suction control	Gentle continuous bubbling in a wet-suction system; vigorous bubbling suggests excessive suction. Maintain prescribed water level.
Water seal	Tidaling with respiration is expected. Continuous bubbling suggests an air leak. Absent tidaling may mean obstruction, dependent loop or lung re-expansion.
Collection	Mark drainage with date/time. Report sudden bright-red output, rapidly increasing output or amounts above facility/provider parameters.

- If the tube leaves the client, apply a sterile occlusive dressing over the site and notify the provider/rapid response per condition.
- If tubing separates from the drainage system, place the tube end in sterile water temporarily according to policy and re-establish a sterile system.
- Keep the drainage system below chest level; do not routinely clamp, strip or milk the tube.

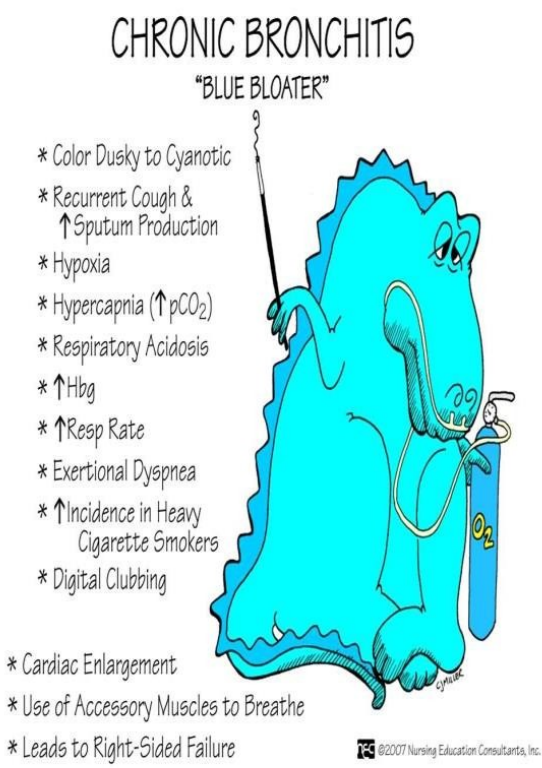
REMOVAL Premedicate as ordered. The client performs Valsalva/holds breath during removal; immediately apply a petroleum-based occlusive dressing.

4. COPD: Emphysema and Chronic Bronchitis

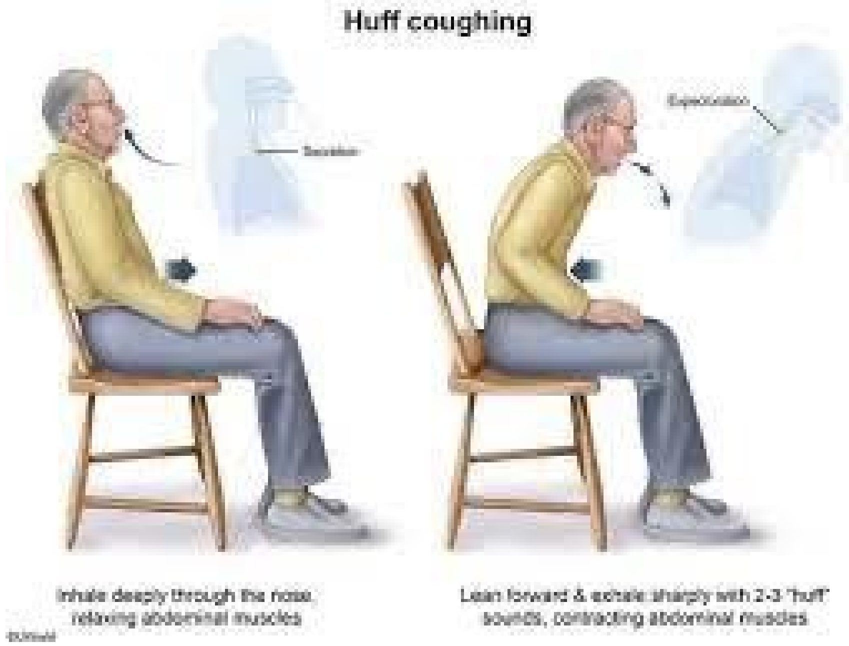
Emphysema	Chronic bronchitis
Alveolar destruction, hyperinflation/barrel chest, thin appearance, minimal cough, dyspnea and pursed-lip breathing. Alpha-1 antitrypsin deficiency may contribute.	Bronchial inflammation and excess mucus; productive cough for at least 3 months in each of 2 consecutive years, edema/cyanosis and recurrent infection.

- Shared findings may include expiratory dyspnea, wheezing, clubbing, polycythemia and cor pulmonale (JVD, hepatomegaly, edema and ascites).
- Teach smoking cessation, pursed-lip/diaphragmatic breathing, huff coughing, vaccination and energy conservation.
- Nutrition: small frequent high-calorie/high-protein meals; rest and use prescribed bronchodilator before meals; avoid gas-forming foods and excess fluids with meals.

Emphysema: classic "pink puffer" comparison image from the original guide



Chronic bronchitis: classic "blue bloater" comparison image from the original guide



Huff-cough technique to mobilize respiratory secretions

5. Respiratory Medications

Class / examples	Key nursing point
Short-acting beta2 agonist: albuterol	Rescue medication for acute bronchospasm; may cause tachycardia, tremor, increased BP or glucose.
Long-acting beta2 agonist: salmeterol	Maintenance, not rescue monotherapy for an acute asthma attack.
Anticholinergic: ipratropium, tiotropium	Bronchodilation; dry mouth, blurred vision, urinary retention and tachycardia may occur.
Inhaled corticosteroid: fluticasone, budesonide	Maintenance anti-inflammatory; rinse mouth after use to reduce oral candidiasis.
Leukotriene modifier: montelukast, zafirlukast	Prevention/maintenance; monitor neuropsychiatric or liver-related concerns as appropriate.
Theophylline/aminophylline	Narrow therapeutic range; monitor level and toxicity such as nausea, tachycardia or seizures.

INHALER ORDER When both are prescribed, use the bronchodilator first, wait as directed, then the inhaled corticosteroid; rinse the mouth after the steroid.

6. Oxygen Delivery Devices

Device	Usual flow / concentration	Priority
Nasal cannula	1-6 L/min; ~24-44%	Allows eating/talking; use water-soluble lubricant and assess ears/nares.
Simple mask	At least 5 L/min; ~35-60%	Minimum flow prevents CO_2 rebreathing; avoid with vomiting and assess claustrophobia.
Venturi mask	Device-specific; precise FiO_2	Best when an exact oxygen concentration is needed; keep entrainment ports open.
Partial rebreather	Usually 6-10 L/min; ~40-70%	Keep reservoir bag partially inflated; bag must not collapse on inspiration.
Nonrebreather	Usually 10-15 L/min; up to ~90-95%	Emergency high concentration; keep bag inflated and one-way valves functional.
Trach collar / T-piece	Prescribed FiO_2	Add humidification, empty condensation away from client, keep exhalation port open.

Nasal cannula

Simple oxygen mask

Venturi mask with color-coded adapters

Partial-rebreather mask

Nonrebreather mask with reservoir and one-way valves

7. Asthma

- Triggers include allergens, smoke/pollution, occupational chemicals, perfume/aerosols, exercise, GERD, aspirin/NSAIDs and nonselective beta blockers.
- Expected findings: chest tightness, expiratory dyspnea, wheezing, cough, nasal flaring and tachypnea.

Peak-flow zone	Meaning / action
Green: ≥80% personal best	Good control; continue prescribed plan.
Yellow: 50-79%	Caution; use rescue medication and follow action plan/provider instructions.
Red: <50%	Medical alert; use rescue medication and seek emergency help according to plan.

Peak-flow meter used to monitor airflow limitation

RED CODE A “silent chest,” exhaustion, altered mental status or decreasing wheeze with worsening distress suggests minimal airflow and impending respiratory failure.

8. Pulmonary Embolism

Cause	Examples / cue
Thromboembolism	DVT risk: immobility, surgery, pregnancy, obesity, long travel and estrogen therapy.
Fat embolism	Long-bone/pelvic fracture; respiratory distress, neurologic change and petechiae often 24-72 hr later.
Air embolism	Central/IV line entry. Prevent air entry; position/treat according to emergency protocol.
Amniotic/septic embolism	Pregnancy-related emergency or infected embolus/endocarditis.

- Sudden dyspnea, tachypnea, tachycardia, pleuritic chest pain, hypoxemia, hemoptysis, syncope or confusion.
- D-dimer helps rule out in low-risk patients; CT pulmonary angiography is commonly definitive when appropriate. V/Q scan is an alternative.
- Provide oxygen, elevate HOB, anticoagulation; thrombolysis or embolectomy may be required for hemodynamic instability.

9. Pneumonia and Ventilator-Associated Pneumonia

Community acquired	Hospital acquired
Begins in community or within first 48 hr of admission; common pathogens include Streptococcus pneumoniae and atypical organisms.	Begins ≥48 hr after admission; consider resistant gram-negative organisms and MRSA based on setting/risk.

- Findings: fever/chills, pleuritic pain, unilateral crackles, dull percussion, increased fremitus, dyspnea and purulent/rust-colored sputum.
- Obtain cultures before antibiotics when possible without delaying urgent care. Give oxygen, antimicrobials, fluids as appropriate, antipyretics, coughing/deep breathing and early mobility.
- VAP bundle: HOB 30-45°, regular oral care, hand hygiene, daily sedation interruption/readiness-to-extubate assessment and appropriate cuff-pressure monitoring.

SPUTUM Pink frothy sputum suggests acute pulmonary edema, not routine pneumonia.

10. Tuberculosis

- Symptoms: persistent cough, night sweats, low-grade fever, fatigue, anorexia/weight loss and possible blood-streaked sputum.
- Airborne precautions: negative-pressure room; staff wear fit-tested N95 or higher respirator. The client wears a surgical mask during transport.
- Testing may include IGRA (QuantiFERON), TST/PPD, chest imaging and three sputum specimens for smear/culture. Prior BCG vaccination can affect TST but not IGRA.

Medication	High-yield adverse effect / teaching
Rifampin	Orange-red body fluids, hepatotoxicity and many drug interactions; reduces hormonal contraceptive effectiveness.
Isoniazid (INH)	Hepatotoxicity and peripheral neuropathy; pyridoxine (vitamin B6) is often prescribed.
Pyrazinamide	Hepatotoxicity and hyperuricemia/gout.
Ethambutol	Optic neuritis; baseline and periodic visual acuity/color testing.
Streptomycin	Ototoxicity and nephrotoxicity.

ADHERENCE Active TB requires multidrug therapy for the full prescribed course. Stopping early promotes drug resistance.

11. ARDS and Other Pulmonary Disorders

Disorder	Key findings / management
ARDS	Acute noncardiogenic pulmonary edema with bilateral infiltrates and refractory hypoxemia after sepsis, shock, aspiration, trauma or other insult. Mechanical ventilation with lung-protective strategy, prone positioning and treatment of cause.
Legionnaires disease	Pneumonia from contaminated aerosolized water; fever, GI/neurologic findings may occur. Treat with prescribed antibiotics; standard precautions.
Histoplasmosis	Fungal spores associated with bird/bat droppings. Antifungal therapy for significant disease; reduce exposure when cleaning contaminated areas.
Sarcoidosis	Multisystem granulomatous disease; cough, dyspnea, constitutional symptoms, skin/eye/joint findings. Corticosteroids may be used.
Occupational lung disease	Silicosis, coal workers pneumoconiosis, asbestosis, talcosis and berylliosis. Remove exposure, use PPE and provide supportive respiratory care.

FINAL NCLEX FOCUS Prioritize airway and oxygenation, recognize silent chest/tension pneumothorax/refractory hypoxemia, assess the client before equipment, and match isolation precautions to the disease.